Mumbi Whidby

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Research Interests

My Ph.D. research focuses on **tactile sensing** and **manipulation** for robotic and human-operated systems. I design and fabricate **tactile sensors**, from microfluidic skins to **wearable haptic devices**, and use them to support robotic manipulation and to restore lost tactile feedback in **degraded-sensing environments**, such as underwater settings. My current work also integrates multimodal tactile sensing into **Vision-Language-Action models**, enabling robots to reason about physical properties across sequential grasps. I aim to advance how both robots and human operators gain usable information from touch, particularly in conditions where vision or natural sensation alone is insufficient.

Education

University of California, Los Angeles — Los Angeles, CA
Ph.D., Mechanical Engineering (Design, Robotics, and Manufacturing) — *Sept. 2022–Present*
M.S., Mechanical Engineering — *March 2024*

Spelman College — Atlanta, GA
B.S., Computer Science, *Magna Cum Laude* — *May 2022*

Technical Skills

Programming: Python, C++, MATLAB
Tools/Frameworks: ROS, ROS2, OpenCV, SolidWorks, Onshape, Arduino, Git
Fabrication: soldering, silicone molding and casting, basic circuit assembly

Research Experience

UCLA Biomechatronics Lab — Graduate Researcher — *Sept. 2022–Present*
Advisor: Prof. Dr. Veronica J. Santos

- Designed and fabricated a tactile sensor skin for robotic fingertips, integrating a microfluidic liquid-metal (EGaIn) taxel developed by University of Washington collaborators, evaluating four sensor skin configurations across core geometry, overmold thickness, and taxel placement. This work was conducted as part of an ONR-funded, multi-institution bimanual teleoperation project.
- Characterized sensor performance through benchtop tests and validated through real-world grasping tests, teleoperating a LEAP hand gripper via MANUS Metagloves Pro and replaying recorded grasp trajectories for repeatable trials.
- Developing a robotic manipulation system built on a Vision-Language-Action model with a custom dual-modality tactile sensor, enabling the robot to infer and compare physical object properties, such as roughness, compliance, and weight, across sequential grasps.
- Designed and built a prototype wearable haptic feedback device to restore lost tactile discrimination for users in protective or sensing-degraded environments, such as underwater settings. Currently developing the experimental protocol and human-subjects study design.
- Mentored six undergraduate researchers across tactile sensor fabrication, testbed development, and VLA-based manipulation research.

Robotic Platform for Diagnostic Flexible Laryngoscopy — Co-Hardware Lead — *Aug. 2025–Present*

- Co-developing a robotic system to automate diagnostic flexible laryngoscopy, in collaboration with otolaryngologists from the UCLA David Geffen School of Medicine. The system autonomously navigates and controls the procedure, and has been validated through cadaver testing.
- Filed an invention disclosure that led to a provisional patent, held jointly with the project team.
- Completed the NSF I-Corps Regional Program, conducting customer discovery to evaluate the technology's commercial viability.
- Placed 1st in the UCLA Lowell Milken Institute-Sandler Prize for New Entrepreneurs, 1st in the UCLA Knapp Venture Competition, and 3rd in the UCLA Easton Technology Management Center Innovation Challenge (Health Tech track).

Human Fusions Institute, Case Western Reserve University — Research Assistant *Oct. 2023–Sept. 2025*
ONR-Funded Bimanual Teleoperation Project

- Evaluated glove-based hand-tracking hardware (Rokoko Smartgloves, MANUS Metagloves Pro) as an alternative input method for the project's teleoperation pipeline.
- Demonstrated a closed-loop system in which the gloves teleoperated a LEAP hand gripper fitted with tactile fingertip sensors, relaying touch feedback to the operator through ACE neural haptic display rings.

NASA L'SPACE Mission Concept Academy — Researcher *Jan.–May 2020*

- Contributed to payload design and engineering analysis as part of a student team developing a mission concept for NASA's Lucy Mission, applying systems engineering principles and NASA mission lifecycle protocols under the guidance of NASA and industry professionals.
- Co-authored sections of the team's Preliminary Design Review (PDR), documenting technical requirements and mission feasibility.

U.S. Army HBCU/MI Drone Design Competition — Engineering Team Member *April 2019*
Sun Bowl Stadium, El Paso, TX

- Contributed to systems integration for a biologically fabricated, approximately 80% biodegradable UAV, incorporating flight control hardware, sensors, and electronics into a bio-based airframe developed by a multidisciplinary biology team, as part of the inaugural U.S. Army HBCU/MI Drone Design Competition.
- Conducted ground and flight testing and remotely piloted the UAV to complete obstacle navigation tasks, performing evaluation and troubleshooting to improve flight stability and handling.

Industry Experience

Honeywell Enterprise — Launch IT Intern *May–Aug 2021*

- Investigated opportunities for robotics integration within large-scale warehouse and logistics systems, including automated storage and retrieval systems (AS/RS) and autonomous mobile robots (AMRs), evaluating sites across the U.S. and U.K. in coordination with teams based in the U.S. and India.
- Designed and implemented AWS-based cloud solutions to support rapid deployment and real-time system monitoring of robotic platforms, reducing deployment time and operational risk.
- Conducted cost–benefit analyses of robotic platforms across multiple fulfillment scenarios to inform investment strategy and lifecycle decisions.

Publications

1. Whidby, M., Kang, S.-M., Torky, J., Lesaicherre, H., Penaloza, J., Bender, A.T., Posner, J.D., and Santos, V.J. "Design Methodology for a Microfluidic Tactile Sensor Skin for Robot Fingertips." In preparation.
2. Harber, E., Johnson, C.P., Liebman, A., Psychoyos, A., Whidby, M., Kang, S.-M., Penaloza, J., Bender, A.T., Posner, J.D., Santos, V.J. "OptiStrain: A Vision- and Microfluidics-based Tactile Sensor with High Spatial and Temporal Resolution." In preparation.

Awards and Honors

Cota-Robles Fellowship. Prestigious, multi-year University of California fellowship awarded to entering Ph.D. students based on academic merit and research potential, with emphasis on supporting students from backgrounds underrepresented in academia.

Boeing Scholar. Competitive, merit-based scholarship recognizing leadership, innovation, and outreach in engineering; received multiple years.

Upsilon Pi Epsilon. Member of the international honor society for the computing and information disciplines, recognizing outstanding academic achievement, endorsed by the ACM and IEEE Computer Society.